

MACHINE LEARNING EXAM CHEAT SHEET

1. DATA PRE-PROCESSING

Garbage In, Garbage Out. Always do this first!

Feature Scaling

Required for distance-based algorithms (k-NN, SVM, Gradient Descent). **Not** required for Trees.

- **Min-Max Normalization:** Scales to $[0, 1]$.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

- **Standardization (Z-score):** Mean 0, Var 1.

$$x' = \frac{x - \mu}{\sigma}$$

Encoding Categorical Data

- **Label Encoding:** Assign integer (0, 1, 2). Use for ordinal data (Low, Med, High). *Risk:* Model assumes $2 > 1$.
- **One-Hot Encoding:** Create binary columns ($[1, 0, 0], [0, 1, 0]$). Use for nominal data (Red, Blue, Green).

Handling Missing Data

- **Drop:** Only if dataset is huge.
- **Impute Mean/Median:** Standard approach.
- **Impute Mode:** For categorical features.

2. OPTIMIZATION: GRADIENT DESCENT

Goal: Minimize Cost Function $J(w)$. Update Rule: $w := w - \alpha \nabla J(w)$.

The Three Variants

1. Batch Gradient Descent:

- Uses **ALL** N training examples for 1 update.
- *Pros:* Stable convergence, optimal path.
- *Cons:* Very slow for large datasets.

2. Stochastic Gradient Descent (SGD):

- Uses **ONE** random example for 1 update.
- *Pros:* Very fast, escapes local minima.
- *Cons:* High variance, "noisy" path.

3. Mini-Batch Gradient Descent:

- Uses a small batch ($b \approx 32 - 256$) for 1 update.
- *Verdict:* The Best of Both Worlds. Standard in Deep Learning.

3. LINEAR LOGISTIC REGRESSION

Linear Regression (Continuous y)

- **Model:** $h_w(x) = w^T x$.
- **Loss (MSE):** $J = \frac{1}{2N} \sum (h - y)^2$.
- **Solution:** $w = (X^T X)^{-1} X^T y$ (Normal Eq).

Logistic Regression (Binary y)

- **Sigmoid:** $\sigma(z) = \frac{1}{1+e^{-z}}$. Squashes to $[0, 1]$.
- **Model:** $P(y = 1|x) = \sigma(w^T x)$.
- **Loss (LogLoss):** $-[y \log(h) + (1 - y) \log(1 - h)]$.
- **Decision Boundary:** $w^T x = 0$.

4. DECISION TREES

Goal: Split data to maximize purity.

Splitting Metrics

1. Entropy (Information Theory):

$$H(S) = - \sum p_i \log_2(p_i)$$

- Pure (all same) $\rightarrow$ Entropy 0.
- Impure (50/50) $\rightarrow$ Entropy 1.

2. Information Gain (IG):

$$IG = H(\text{Parent}) - \sum \text{Weighted_Avg} \cdot H(\text{Children})$$

3. Gini Impurity: $1 - \sum p_i^2$ (Computationally faster).

Overfitting vs. Underfitting in Trees

• Overfitting (High Variance):

- *Symptoms:* Tree is very deep, leaf nodes have 1 sample, perfect training accuracy but poor test.
- *Fix:* ****Pruning**** (Pre-pruning: `max_depth`, `min_samples_split`; Post-pruning).

• Underfitting (High Bias):

- *Symptoms:* Tree is too shallow (stump), captures no patterns.
- *Fix:* Increase depth, add more features.

5. EVALUATION METRICS

Confusion Matrix

	Pred Positive	Pred Negative
Actual Pos	TP	FN (Type II)
Actual Neg	FP (Type I)	TN

Key Formulas

- **Accuracy:** $\frac{TP+TN}{Total}$ (Bad for imbalanced data).
- **Precision:** $\frac{TP}{TP+FP}$ (Trust result?).
- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$ (Find all?).
- **F1-Score:** $2 \cdot \frac{P \cdot R}{P+R}$ (Balance).

AUC - ROC Curve

Performance across ALL classification thresholds.

- **X-Axis:** FPR (False Positive Rate) = $\frac{FP}{FP+TN}$.

- **Y-Axis:** TPR (Recall/Sensitivity) = $\frac{TP}{TP+FN}$.

- **Curve:** Plots TPR vs FPR at different thresholds (0.1, 0.5, 0.9...).

• AUC (Area Under Curve):

- $AUC = 0.5 \rightarrow$ Random guessing.
- $AUC = 1.0 \rightarrow$ Perfect classifier.
- Higher AUC means better ability to distinguish classes.

6. INSTANCE BASED (K-NN)

- **k=1:** High Variance (Sensitive to outliers).
- **k=N:** High Bias (Predicts majority class).
- **Curse of Dimensionality:** As features increase, distance becomes meaningless.

7. SUPER AWESOME EXAM STRATEGY

1. Which Algorithm?

- Small data, Interpretability needed? → **Decision Tree**.
- High dimensions, linear boundary? → **Logistic Reg.**
- Complex non-linear? → **k-NN**.

2. Bias-Variance Hack:

- Train Error $\approx$ Test Error (both high) → **Bias**.
- Train Error $\ll$ Test Error → **Variance (Overfitting)**.

3. Calc Shortcuts:

- $\log_2(1) = 0, \log_2(0.5) = -1$.
- For Entropy, if $p_+ = 0.5$, Entropy is 1.

4. **Pre-Processing Trap:** If question asks about k-NN or Gradient Descent, check if features are scaled. If not, deduct marks or mention normalization!